A FORMAL ALGEBRAIC SPACE IS AFFINE EXACTLY WHEN ITS REDUCTION IS AFFINE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof refers to the structure theorem as 'Yasuda's lemma'; that reference is carried as a navigation stub and the statement is recorded as a standard prerequisite.

Editorial note. The counted spans carry 8 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The characterisations of the quasi-separated and quasi-compact types of a formal algebraic space. Same chapter, Lemmas lemma-characterize-quasi-separated and lemma-characterize-quasi-compact, invoked at proof lines 4026-4028. Give that the space is quasi-compact and quasi-separated.
- The structure theorem for quasi-compact quasi-separated formal algebraic spaces. Same chapter, Lemma lemma-structure-quasi-compact-quasi-separated, invoked at proof lines 4029-4032. Produces the presentation as a filtered colimit of thickenings of algebraic spaces.
- A thickening of an affine algebraic space is affine. Limits of Spaces, Lemma spaces-limits-lemma-reduction-scheme, invoked at proof lines 4032-4036. Shows the members of the presentation are affine.

Source reference. Original definition 0AJA: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 1967–1967. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original definition 0AJ7: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 601–601. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AJ9: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 3776–3776. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:formal-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0AJ5: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 3580–3580. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AIN: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2833–2833. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AJE: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 3866–3866. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 07VU: sources/stacks-chow/tags.txt, source lines 10019–10019. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let S be a scheme. Let X be a formal algebraic space over S. Then X is an affine formal algebraic space if and only if its reduction X_{red} (Lemma 0AIN) is affine.*

Proof. By Lemmas 0AJ5 and 0AJ9 and Definitions 0AJ7 and 0AJA we see that X is quasi-compact and quasi-separated. By Yasuda's lemma (Lemma 0AJE) we can write $X = \operatorname{colim} X_\lambda$ as a filtered colimit of thickenings of algebraic spaces. However, each X_λ is affine by Limits of Spaces, Lemma 07VU because $(X_\lambda)_{red} = X_{red}$. Hence X is an affine formal algebraic space by definition. □